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6 **ABSTRACT**

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8 **Introduction**

9 **Model**

Table 1. Model parameters.

Symbol	Description	Unit	Experimental litterature
R_0	Recycling pool capacity	-	5-20 vesicles per bouton in cultured rat hippocampal neurons ^{1,2}
D_0	Docked pool capacity	-	
$\tau_{R \rightarrow D}$	Vesicle transition rate from recycling to docked		
$\tau_{D \rightarrow R}$	Vesicle transition rate from docked to recycling (mixing)		
$\tau_{F \rightarrow R}$	Vesicle transition rate from fused to recycling		
Calcium stuff			

10 **References**

11 **1.** Rizzoli, S. O. & Betz, W. J. Synaptic vesicle pools. *Nat. Rev. Neurosci.* **6**, 57–69, [10.1038/nrn1583](#) (2005).
12 **2.** Schikorski, T. & Stevens, C. F. Morphological correlates of functionally defined synaptic vesicle populations. *Nat. Neurosci.*
13 **4**, 391–395, [10.1038/86042](#) (2001).